

MIXED-MODE FRACTURE IN BIAXIAL STRESS STATE: APPLICATION OF THE DIAMETRAL-COMPRESSION (BRAZILIAN DISK) TEST

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Abstract—Mixed-mode fracture of soda-lime glass was studied using a diametral-compression test that features disk specimens with symmetric through-cracks. The test enables one to study fracture under pure mode I loading, pure mode II loading, or any combination of mode I and mode II loading by a simple alignment of the crack relative to the diameter of compression loading. The disk specimens were precracked with the aid of both chevron notches and water-assisted subcritical crack growth. The directions of noncoplanar crack extensions and the relative magnitudes of mode I and mode II stress-intensity factors for mixed-mode fracture under inert conditions were compared to the predictions of three different mixed-mode fracture theories. None of the theories was completely adequate to explain the experimental observations, but a maximum hoop stress criterion modified to include second order, nonsingular term in the series solution for the crack-tip region stress gave reasonable agreement with the experimental results.

1. INTRODUCTION

FRACTURE of brittle materials from inclined cracks that are subject to both remote tension and shear loading, i.e. the mixed-mode fracture problem in fracture mechanics, was first studied by Erdogan and Sih in 1963[1]. Their experiments included fracture from inclined cracks in PMMA plates subjected to uniform tension (mode I and mode II loading) and fracture from straight through-cracks subjected to equal and opposite concentrated shear loads applied to the crack surface (pure mode II loading). An analysis based on maximum crack-tip hoop tension was found to explain satisfactorily both the direction of crack extension as well as the critical loads. Panasyuk and coworkers[2] reported a similar study on inclined cracks in both inorganic and organic glasses in 1964 and they also analyzed their results in terms of the hoop stress criterion.

Williams and Ewing[3] subsequently conducted more extensive experiments on thin plates of PMMA with cracks set at various angles to the applied uniaxial stress. They showed that inclusion of the nonsingular stress component parallel to the crack in their analysis of the critical hoop stress at a critical distance from the crack tip significantly improved the correlation between the theory and the experiments. Finnie and Saith[4] corrected the analysis of Williams and Ewing and showed further improvement in the agreement between the hoop stress theory and the experimental results of Williams and Ewing.

In the years following these early studies there have been many new developments in both the theories and the experimental techniques used to study mixed-mode fracture. Sih[5] has advanced strain energy density criterion for mixed-mode crack problems. Strain energy release rate analyses for noncoplanar crack extension in mixed-mode fracture have been formulated by Hussain, Pu and Underwood[6], Palaniswamy and Knauss[7], Chiang[8] and Ichikawa and Tanaka[9]. Despite the distinctly different physical criteria invoked in the hoop stress, the strain energy density and the strain energy release rate theories, their predictions with respect to the crack-extension directions and the critical fracture loads are remarkably close. Ewing, Swedlow and Williams[10] and Swedlow[11] compared their experimental results with the predictions of the critical hoop stress theory and Sih's strain energy density theory and found little basis to discriminate between the two criteria. Similarly, Palaniswamy and Knauss[7] found very similar predictions for their strain energy release rate theory when compared to the hoop stress theory.

In contrast to the many theoretical developments, studies devoted to experiments in mixed-mode fracture have been surprisingly limited. In addition to the experiments cited above, Palaniswamy and Knauss[7] and more recently Kordisch[12] have also conducted experiments with inclined cracks in brittle polymers and inorganic glasses, respectively. Experiments involving inclined cracks in plates subjected to remote uniaxial tension or bending, however, have one limitation. Crack loadings that produce pure mode II stress state cannot be achieved in these tests. To overcome this limitation a number of investigators have designed pure mode II test specimens[13, 14] or used notched round bar specimens in combined tension/torsion tests[15, 16]. But the test specimens are generally complex in design and difficult to fabricate from brittle materials.

The center-cracked diametral-compression disk specimen provides a simple means for overcoming these problems. Mixed-mode stress states ranging from pure mode I to pure mode II can be achieved in this specimen by selecting the angle of inclination of the central through-crack relative to the diametral line of compression loading. Yarema and coworkers[17, 18] employed this test specimen to measure mode I fracture toughness of concrete and mode I and mixed-mode I and mode II fracture toughness of a sintered carbide. Awaji and Sato[19] used the test to study mixed-mode fracture of graphite, plaster and marble. But both of these studies employed disk specimens with machined notches rather than precracked disk specimens. Analytical solutions for mode I and mode II stress-intensity factors for through-cracks in the diametral-compression specimen have been reported by Libatskii and Kovchik[20] and Atkinson and coworkers[21]. The latter group also reported some limited data for mixed-mode fracture toughness of PMMA.

A number of studies in recent years have investigated mixed-mode fracture of structural ceramics[16, 22–24]. With the exception of the combined tension/torsion experiments of Petrovic[16] on notched round test bars, these studies have used inclined surface flaws produced by microhardness indentation in bend or biaxial tension–compression test specimens[22–24]. The indentation technique has been used in ceramics because of the ease of precracking of the test specimens. Analysis of the surface-flaw fracture data in mixed-mode loading is, however, complicated by the three dimensional crack geometry and the effects of the residual stress of the indentation plastic zone. As a result, agreement between the experiments and the mixed-mode fracture theories is poor for the surface-flaw tests[24].

The limitations of the surface-flaw technique led to the present development and application of the center-cracked diametral-compression specimen for a fundamental study of mixed-mode fracture. The present authors have developed a novel precracking technique for the diametral-compression specimen using a chevron notch. The technique enables study of mixed-mode fracture in both model brittle materials such as inorganic glasses as well as polycrystalline ceramics using natural sharp cracks rather than machined notches. Precracked disk specimens have also been used by the authors to measure mode I fracture toughness of ceramics[25]. The present study was undertaken to determine if the improved disk test technique could aid in distinguishing among the different mixed-mode fracture theories. The experiments were conducted on soda-lime glass, a model brittle material, in order to avoid the microstructural effects encountered in structural ceramics or the material nonlinearity effects associated with the tensile yielding at the crack tip in PMMA.

DIAMETRAL-COMPRESSION TEST

The geometry of the chevron-notched disk specimens used in this study is illustrated in Fig. 1. The disk specimens were 25 mm in radius (R) and 2.3 mm in thickness (B). Chevron notches were machined in the center of the disks with a 11 mm radius diamond-grit cutting blade such that the radial extension of the notch on the surface, a_1 , was 6.35 mm. The resulting base length of the notch, a_0 , was 4 mm. The slot opening, N , was $\sim 250 \mu\text{m}$.

The test material used in this study was well-annealed soda-lime glass. Its composition and mechanical properties, including fracture toughness, hardness and elastic properties have been reported in our earlier papers[25, 26].

Soda-lime glass disk specimens were precracked by loading the disk specimens in compression along the diameter parallel to the chevron notch. Drops of deionized water were placed in the notch to promote subcritical crack growth during precracking. Cracks initiated at the chevron-

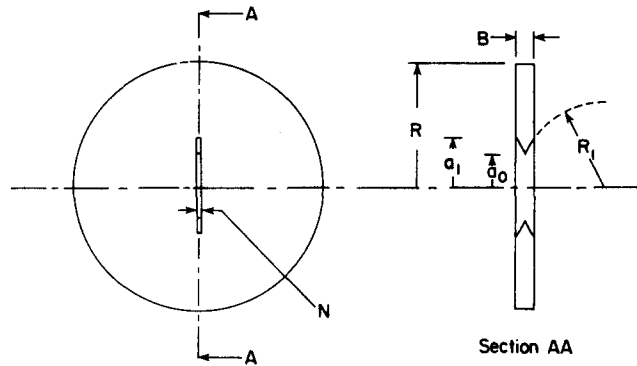


Fig. 1. Geometry of the chevron-notched diametral-compression specimens used in the mixed-mode fracture tests.

notch tip were allowed to grow stably through the chevron-notch section until the crack-front width equalled the disk thickness; i.e. the crack length was equal to a_1 . Following precracking the disk specimens were dried in a low temperature oven at 150°C. Figure 2 shows cracks in the chevron-notch sections of a disk specimen of soda-lime glass. The stability of the cracks in the chevron section is consistent with a compliance analysis developed by the authors in connection with the application of this test geometry for measuring mode I fracture toughness[25].

The mixed-mode fracture test using the precracked disk specimens consisted of loading the disks in compression along a diameter inclined at an angle $(90 - \alpha)$ with respect to the notch plane (see Fig. 3). The angle of inclination α was varied from 90 to 60 degrees at intervals of 2.5 degrees. The tests were conducted in a dry N_2 environment to minimize effects of moisture-assisted slow-crack growth. Figure 4 shows the fracture pattern in a disk specimen loaded at an angle $\alpha = 62.5$ degrees.

Stress-intensity factors for the inclined cracks in the diametral-compression specimens were calculated by using the solutions given by Atkinson and coworkers[21]. Mode I and mode II stress-intensity factors are given by the relations:

$$K_I = \{(Pa^{1/2})/[\pi^{1/2}RB]\} N_I \quad (1)$$

and

$$K_{II} = \{(Pa^{1/2})/[\pi^{1/2}RB]\} N_{II} \quad (2)$$

where P is the load applied in compression, a is half the crack length and N_I and N_{II} are nondimensional coefficients that are functions of the relative crack size, (a/R) , and the angle of inclination of the crack relative to the line of loading. Atkinson and coworkers developed solutions for N_I and N_{II} in the form of series and gave numerical solutions for cracks in the size range $(a/R) = 0.1-0.6$ using a five term approximation. Their solutions were used in the present study by fitting cubic polynomials to the discrete values for purposes of interpolation.

For the disk specimens used in the present study ($a/R = 0.25$), the mode I stress-intensity factor decreased monotonically with decreasing angle α while the mode II stress-intensity factor increased. A pure mode II loading was achieved at an angle between 62.5 and 60 degrees. For the 60 degrees orientation mode I stress-intensity factor was negative.

In addition to the stress-intensity factors the direction of crack extension is also of interest in mixed-mode fracture. Crack extension was non-coplanar with the initial crack for all situations involving mode I and mode II loading. Figure 5 shows the crack-extension pattern in the vicinity of the initial crack tip in a soda-lime glass specimen fractured under mixed-mode loading conditions. Cracks extended abruptly at an initial extension angle, θ^* , that depended on the initial orientation angle, α , and gradually curved to a direction normal to maximum remote tension. The initial direction of crack extension, θ^* , was measured by using a shadowgraph.

DISK TEST RESULTS

Figure 6 summarizes the initial direction of extension of cracks in disk specimens of soda-lime glass as a function of their orientation, i.e. as a function of the relative combination of mode

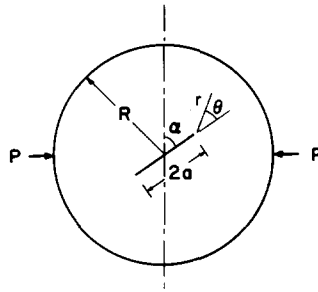


Fig. 3. Inclined crack in a disk specimen and the associated coordinate system.

I and mode II stress-intensity factors. The data points and the error bars represent mean and standard deviations of eight measurements on each disk specimen corresponding to the eight crack edges near the original crack tips. The initial direction of crack extension, θ^* , increased from 0 degrees for pure mode I loading to approximately -65 degrees for pure mode II loading. The dashed line in the figure corresponds to the direction normal to remote maximum tension:

$$\theta^* = -(90 - \alpha). \quad (3)$$

Small surface cracks subjected to mixed-mode loading in the disk test follow the direction defined by eq. (3)[24]. It is clear from Fig. 6 that the behavior of through-cracks is significantly different. The angle of initial extension of through cracks is much greater than that given by eq. (3), especially for stress states approaching pure mode II loading.

Figure 7 shows a plot of mode I and mode II stress-intensity factors for mixed-mode fracture in the disk tests as calculated from the fracture loads using eqs (1) and (2). The critical stress-intensity factor for pure mode I loading, K_{Ic} , was $0.73 \text{ MPa } m^{1/2}$. This value agrees with those measured by other fracture mechanics techniques[25]. With increasing degree of mode II loading, the mode I stress-intensity factor at fracture decreased monotonically. The critical mode II stress-intensity factor, K_{IIc} , for pure mode II loading was $0.9 \text{ MPa } m^{1/2}$. The ratio of the critical values of mode I and mode II stress-intensity factors, $K_{IIc}/K_{Ic} = 1.23$, is in good agreement with similar results reported by Yarema and coworkers[17, 18] and Awaji and Sato[19] for other materials.

Also shown in Fig. 7 are two data points which represent crack orientations for which the mode I stress-intensity factor was negative. The mode II stress-intensity factors corresponding to these two tests are significantly higher, i.e. there is a discontinuity in the slope of the $K_I - K_{II}$ data trend at $K_I = 0$. This result is significant and logical because for negative values of mode I stress-intensity factors the crack faces come into contact with each other and the applied mode II stress-intensity must overcome the frictional resistance of the contacting crack faces.

ANALYSIS OF MIXED-MODE FRACTURE RESULTS

Three different theoretical hypotheses have been advanced to account for mixed-mode fracture observations in brittle materials:

- (a) Maximum Hoop Stress theory
- (b) Minimum Strain Energy Density theory
- (c) Strain Energy Release Rate theory

(a) Maximum Hoop Stress theory

According to this theory a crack subjected to both mode I and mode II loading extends in a direction θ^* along which the crack-tip hoop stress, $\sigma_{\theta\theta}$, is maximum and fracture occurs when the hoop stress attains a material-characteristic critical value σ^* . In the vicinity of the tip of a crack subjected to both mode I and mode II loading the hoop stress is given by the following relation[11]:

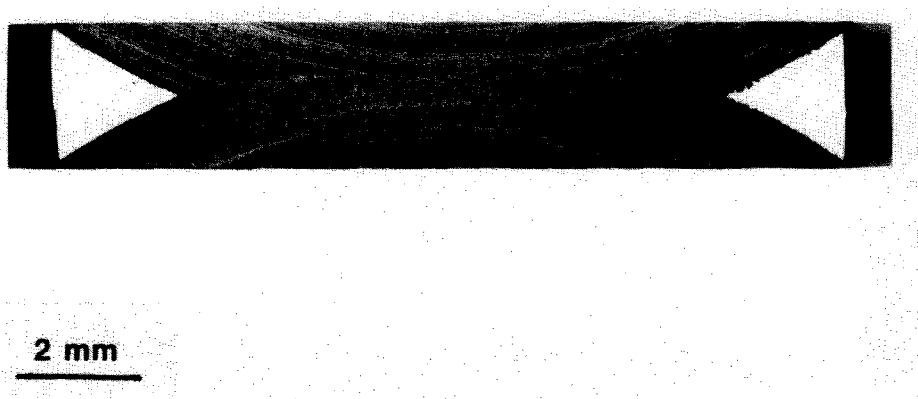


Fig. 2. Stable cracks in the chevron-notch sections of the diametral-compression specimens of soda-lime glass.

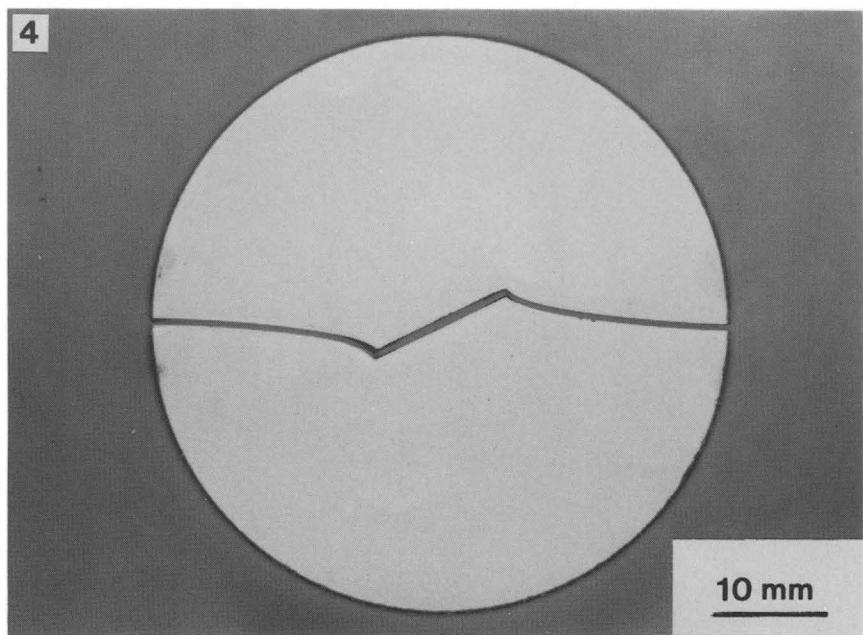


Fig. 4. Fracture pattern in a soda-lime glass disk specimen subjected to mixed-mode loading.

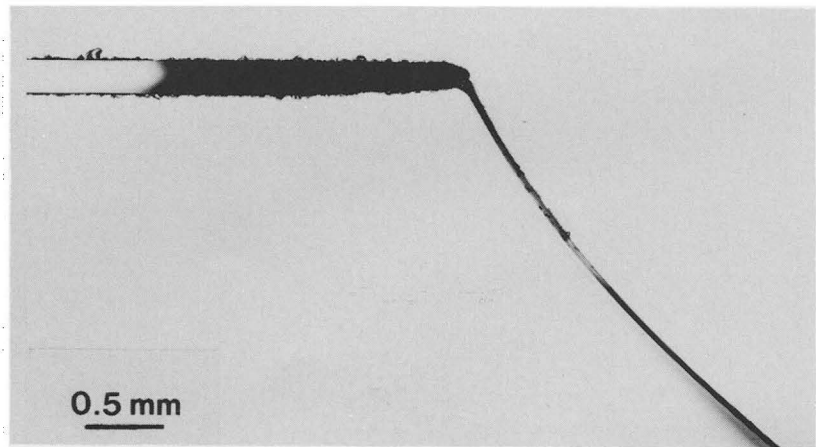


Fig. 5. Non-coplanar crack extension in the vicinity of the original crack tip of an inclined crack in a disk specimen.

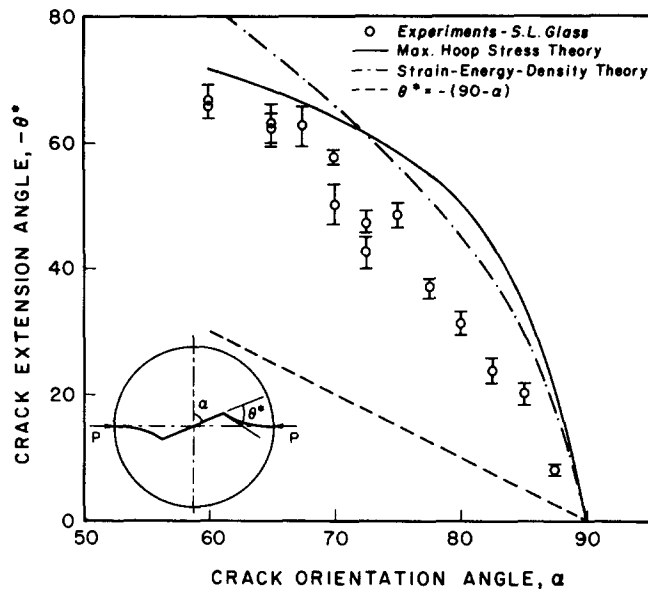


Fig. 6. Direction of extension of cracks in disk specimens subjected to mixed-mode loading.

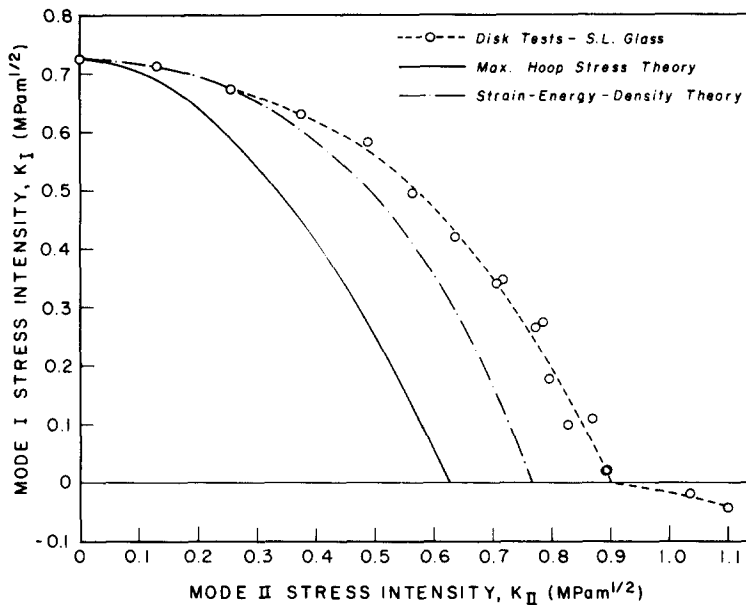


Fig. 7. Mode I and mode II stress intensity factors for mixed-mode fracture in disk tests of soda-lime glass.

$$\sigma_{\theta\theta} = K_I [3 \cos(\theta/2)/4 + \cos(3\theta/2)/4] - K_{II} [3 \sin(\theta/2)/4 + 3 \sin(3\theta/2)/4]. \quad (4)$$

The hoop stress criterion can be stated as follows:

$$\{d\sigma_{\theta\theta}(\theta)/d\theta\} = 0 \quad \text{and} \quad (d^2\sigma_{\theta\theta}/d\theta^2) < 0 \quad \text{for} \quad \theta = \theta^*. \quad (5)$$

Substitution of eq. (4) in eq. (5) provides the crack-extension direction, θ^* , which can then be substituted back in eq. (4) to calculate the mode I and mode II stress intensities that generate the same crack-tip hoop stress as in the case of pure mode I loading. Results of these calculations

are shown by the solid lines in Figs 6 and 7. The hoop stress theory gives reasonable predictions of θ^* for stress states approaching pure mode II, i.e. $\alpha \sim 60$ degrees. But at intermediate values of α the theory overestimates θ^* . In terms of the $K_I - K_{II}$ plot of Fig. 7 the hoop stress theory clearly overestimates the decrease in the critical K_I value due to mode II loading. The hoop stress theory prediction of K_{IIc} for soda-lime glass based on the K_{Ic} value of $0.73 \text{ MPa } m^{1/2}$ is about $0.64 \text{ MPa } m^{1/2}$, which is significantly lower than the experimentally observed value.

(b) *Minimum Strain Energy Density theory*

Sih[5] proposed a criterion for mixed-mode fracture which, in effect, states that a crack subjected to both mode I and mode II loading extends in a direction along which the strain energy density, ϕ , is a minimum and fracture occurs when its value reaches a material-characteristic critical value ϕ^* . In the vicinity of the tip of a crack subjected to both mode I and mode II loading the strain energy density is given by the following relation[5]:

$$\phi = a_{11}K_I^2/r + 2a_{12}K_I K_{II}/r + a_{22}K_{II}^2/r \quad (6)$$

where the coefficients a_{11} , a_{12} and a_{22} are given by the following relations:

$$a_{11} = [(3 - 4\nu - \cos \theta)(1 + \cos \theta)]/(16\pi G) \quad (7)$$

$$a_{12} = [\cos \theta - (1 - 2\nu)] \sin \theta / (8\pi G) \quad (8)$$

$$a_{22} = [4(1 - \nu)(1 - \cos \theta) + (1 + \cos \theta)(3 \cos \theta - 1)]/(16\pi G) \quad (9)$$

where G is the shear modulus and ν is the Poisson's ratio. The minimum strain energy density criterion is stated as

$$\{d\phi(\theta)/d\theta\} = 0 \quad \text{and} \quad (d^2\phi/d\theta^2) > 0 \quad \text{for} \quad \theta = \theta^*. \quad (10)$$

The strain energy density criterion was applied to the disk test results in a manner similar to the hoop stress criterion. The direction of crack extension was first obtained by substituting the strain energy density relations, eqs (6)–(9), in eq. (10). The results so obtained are shown in Fig. 6 by the dot-dash line. The strain energy density criterion also overestimates the angles of crack extension in comparison with the disk test results. The crack extension directions calculated from eq. (10) were substituted back in eqs (6)–(9) to predict the $K_I - K_{II}$ combinations that produce the same strain energy density as in pure mode I loading. The results are shown by the dot-dash line in Figure 7. The theoretical predictions are better than those of the hoop stress criterion, but there is still some discrepancy. Thus, for example, K_{IIc} value predicted by the strain energy density theory is approximately $0.77 \text{ MPa } m^{1/2}$ as compared to the experimental value of $0.9 \text{ MPa } m^{1/2}$.

(c) *Strain Energy Release Rate theory*

Griffith energy balance analyses of nonplanar extension of inclined cracks have been reported by several investigators[6–9]. The analysis of Palaniswamy and Knauss[7] appears to be the most rigorous and their results can be fitted to a simple empirical mixed-mode fracture criterion:

$$K_I/K_{Ic} + 1.5(K_{II}/K_{Ic})^2 = 1. \quad (11)$$

The $K_I - K_{II}$ envelope predicted by eq. (11) is very close to the predictions of the maximum hoop stress criterion. For example, for pure mode II loading eq. (11) gives $K_{IIc} = 0.82 K_{Ic}$ or $K_{IIc} = 0.6 \text{ MPa } m^{1/2}$. Similarly, the crack-extension directions predicted by the strain energy release rate theory is also close to the predictions of the maximum hoop stress theory. Therefore, the experimental results on the disk tests were not separately compared to the strain energy release rate theory predictions.

BIAXIAL STRESS STATE EFFECTS ON MIXED-MODE FRACTURE

Although the results of the disk tests with the through-cracks show significantly better agreement with the mixed-mode fracture theories than do the surface crack results[24] (as seen, for example, in Fig. 6), the systematic deviations observed in Figs 6 and 7 suggested a possible influence of the biaxial stress state in the disk specimen geometry. To illustrate this influence of the stress state the results of the disk tests are plotted in Figs 8 and 9 along with the results of the inclined cracks in uniaxial tension tests of soda-lime glass reported by Panasyuk *et al.*[2] and Kordisch *et al.*[12].

In Fig. 8 the crack-extension directions for inclined cracks, θ^* , measured in both the disk tests and the uniaxial tension tests are plotted versus $[K_{II}/(K_I + K_{II})]$. This parameter was chosen instead of the more common crack-inclination angle α because at the same crack-inclination angle in the two tests the relative magnitudes of the mode I and the mode II stress intensities are different. If the crack-extension angle in mixed-mode fracture is uniquely defined by the relative values of K_I and K_{II} , then the two tests should give identical results when plotted in the manner of Fig. 8. The results plotted in Fig. 8 show, however, that for the same value of (K_{II}/K_I) the measured crack-extension angles are systematically higher in the uniaxial tension tests than in the disk tests.

Figure 9 shows the normalized mode I and mode II stress intensities for fracture from inclined cracks in the diametral-compression and the uniaxial tension tests. Results plotted in Fig. 9 show that the mixed-mode fracture 'envelopes' obtained from the two tests are significantly different. In other words, mixed-mode fracture in soda-lime glass cannot be uniquely defined by mode I and mode II stress-intensity factors.

Williams and Ewing[3] and Finnie and Saith[4] and, more recently, Eftis and coworkers[27, 28] have pointed out that in analyzing mixed-mode fracture from inclined cracks it is not sufficient to express the near crack-tip stresses in terms of only the singular terms, i.e. K_I and K_{II} . By retaining the second order nonsingular stress term in the Williams series solution for a crack in a plate subjected to tension, Williams and Ewing[3] were able to get much better agreement between their inclined crack results and the predictions of the maximum hoop stress criterion. This argument appeared to be relevant to the differences between the diametral-compression test results and the uniaxial tension test results because the nonsingular second term in the expression for the hoop stress includes the effect of a biaxial stress component parallel to the crack. Thus, the effect of the biaxial stress state on mixed-mode fracture can be examined by examining the effect of the second order term.

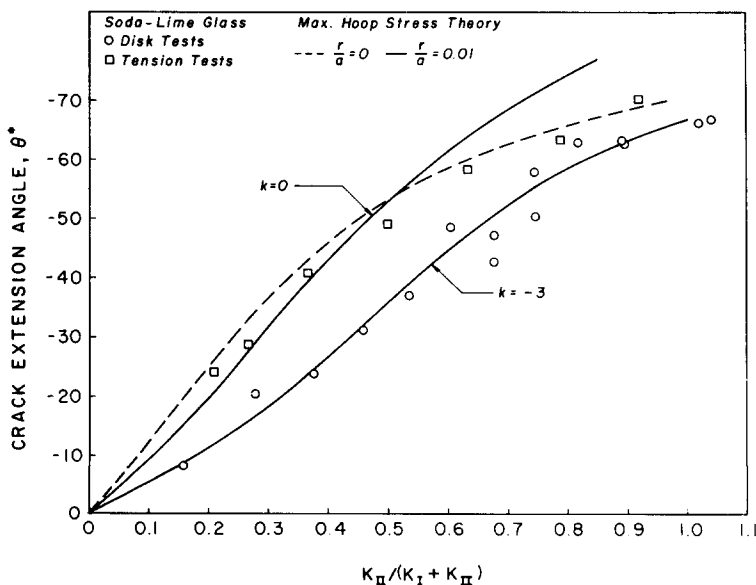


Fig. 8. Crack extension angles for inclined cracks in diametral-compression and uniaxial tension tests as a function of the relative mode I and mode II stress intensities compared with the predictions of the modified hoop stress criterion (for $r/a = 0$ the result is independent of k).

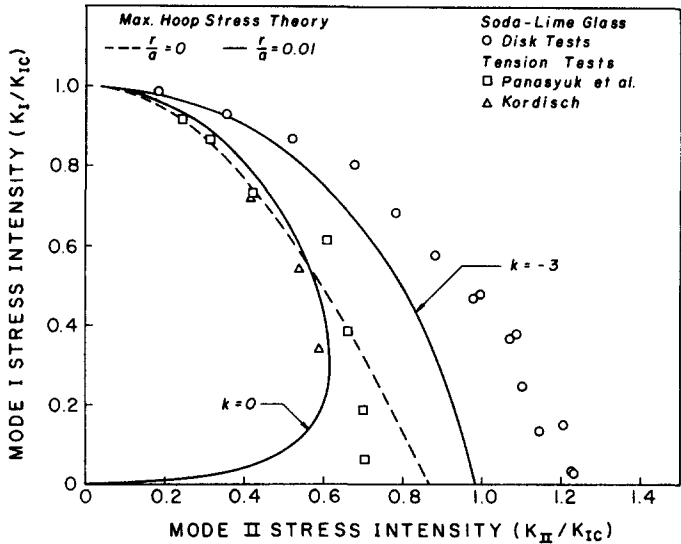


Fig. 9. Normalized mode I and mode II stress-intensity factors for mixed-mode fracture from inclined cracks in the disk and the uniaxial tests of soda-lime glass compared with the predictions of the modified hoop stress criterion (for $r/a = 0$ the result is independent of k).

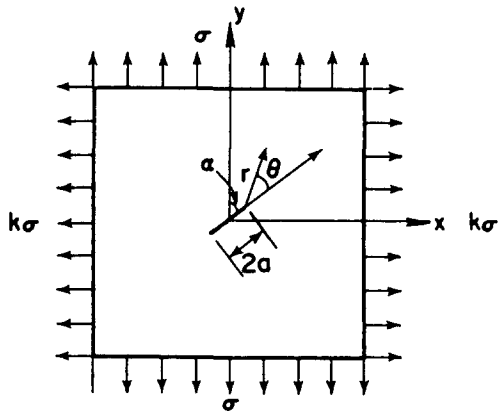


Fig. 10. Inclined crack in a general biaxial stress state.

A precise analysis of the diametral-compression test results along the lines suggested above is not possible because closed-form solution for the second order stress term for the crack in the disk specimen is not available. However, for the limiting case of a short crack in a disk specimen ($a/R \ll 1$) the stress state in the disk specimen should approximate to that of an inclined crack in a plate subjected to a uniform biaxial stress state as shown in Fig. 10. Further, from the stress analysis of the uncracked disk specimen it is known that the biaxial stress state parameter k takes the value $k = -3$ for short cracks. The general representation of the inclined crack in the biaxial stress field, Fig. 10, can also be used to represent the uniaxial tension case because the stress state parameter k takes the value of 0 for this situation.

Eftis and Subramonian[27] have formulated several crack-tip stress field parameters for the crack geometry and stress state shown in Fig. 10 by explicitly considering the second order nonsingular stress term. For example, the crack-tip hoop stress for the inclined crack in a general biaxial stress state is given as[27]

$$\begin{aligned} \sigma_{\theta\theta} = & K_I[3 \cos (\theta/2)/4 + \cos (3\theta/2)/4] - K_{II}[3 \sin (\theta/2)/4 + 3 \sin (\theta/2)/4] \\ & + \sigma(1-k) \cos 2\alpha \sin^2 \theta \end{aligned} \tag{12}$$

where the mode I and mode II stress-intensity factors are given by the following general relations:

$$K_I = \{\sigma(\pi a)^{1/2}/2\}[(1+k) - (1-k) \cos 2\alpha] \quad (13)$$

$$K_{II} = \{\sigma(\pi a)^{1/2}/2\}(1-k) \sin 2\alpha. \quad (14)$$

In applying the short-crack approximation of Fig. 10 to the diametral-compression specimen the remote uniform applied stress σ is related to the load applied in compression through the relation

$$\sigma = (P/\pi RB). \quad (15)$$

The effect of the biaxial stress state on the mixed-mode fracture of an inclined crack can now be examined by employing the maximum hoop stress criterion (eq. 5) with the hoop stress defined by eq. (12). The presence of the nonsingular term in eq. (12), however, now requires that the hoop stress be evaluated at a location near the crack tip that is a small but finite distance removed from the tip. The normalized location parameter should be ideally chosen on the basis of physical considerations such as a characteristic microstructural dimension or the size of a process-zone. In a model brittle material such as soda-lime glass it is not clear what the physical implications of such a location parameter might be. A crack-tip location of $(r/a) = 0.01$ was arbitrarily chosen to assess the effect of the biaxial stress state through the influence of the second order term in eq. (12).

The solid lines in Figs 8 and 9 are the predictions of the maximum hoop stress theory with the effect of the biaxial stress state included into the formulation via the nonsingular term. The predictions for the disk tests and the uniaxial tests have been identified by the stress state parameters, $k = -3$ and $k = 0$, respectively.

Figure 8 shows that the modified hoop stress analysis gives better predictions. In fact, the prediction for the disk tests shows excellent agreement with the experimental results despite the short-crack approximation and the arbitrary value chosen for (r/a) . The agreement with the tension test results in Fig. 8 is not as good, the hoop stress analysis with only the singular terms (shown by the dashed line in Fig. 8) gives as good or even better prediction than the modified hoop stress theory.

Figure 9 compares the normalized $K_I - K_{II}$ fracture envelopes predicted for the two tests with the respective test data. The predicted fracture envelope for the uniaxial test ($k = 0$) does not intersect the K_{II} axis because as α approaches 0 degrees both mode I and mode II stress intensities tend to zero (see eqs 13 and 14) and the local hoop stress is largely determined by the nonsingular term. The agreement between the predicted fracture envelopes and the experimental results is not very good, although the inclusion of the second order term does qualitatively explain the correct trend in the $K_I - K_{II}$ data for the two tests. But the hoop stress criterion underpredicts the values of K_{II} for the disk tests. Uncertainties in selecting the location parameter (r/a) and the stress state parameter k for the disk test contribute to the difficulty in making a precise quantitative comparison.

In principle, analysis of mixed-mode fracture in terms of the crack-tip stress solutions modified to include the second order stress terms can be done by employing the other two fracture criteria, viz. the strain energy density criterion and the strain energy release rate criterion, although these analyses get increasingly complex. The strain energy density criterion was investigated to see if it can give better predictions than the hoop stress criterion. The method of calculations and the results are discussed in the Appendix. In summary, the strain energy density criterion gave results comparable to the hoop stress criterion when the same values are used for the parameters (r/a) and k in the two analyses.

DISCUSSION

The results of this experimental investigation support two conclusions. First, mixed-mode fracture in brittle materials cannot be uniquely defined by the stress-intensity factors. The results of the inclined crack fracture experiments using the diametral-compression tests are significantly different from the uniaxial tension data, especially in the values of the mode II stress intensities at fracture; these differences cannot be attributed to experimental scatter. To illustrate this point

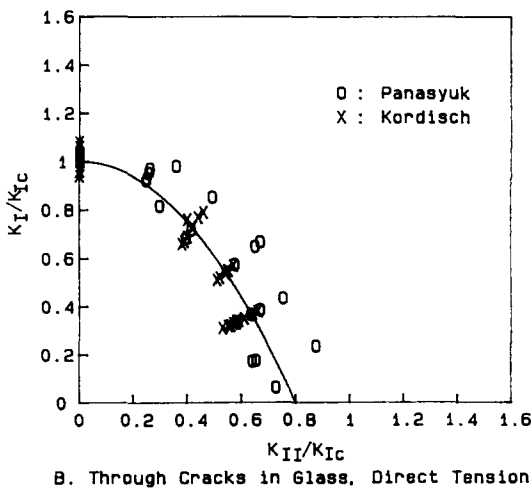
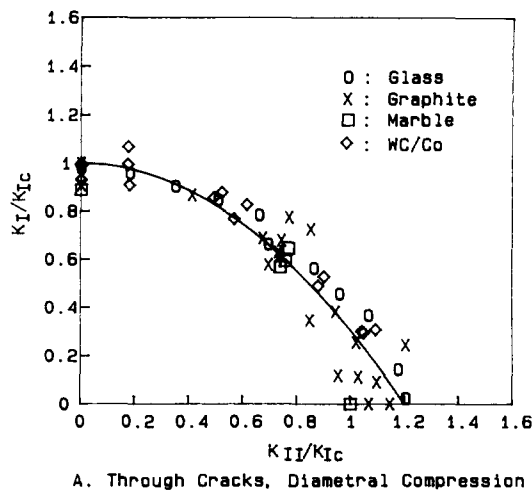
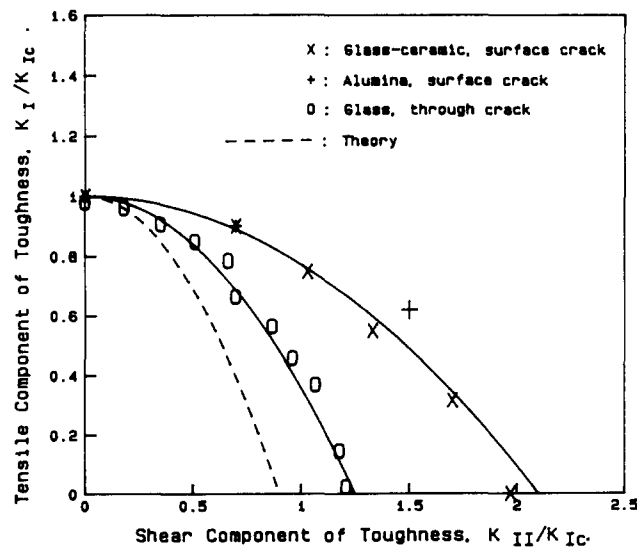


Fig. 11. Normalized mode I and mode II stress-intensity factors for mixed-mode fracture of ceramics in uniaxial tension and diametral-compression tests.



further, data reported in the literature for these two types of tests on a variety of brittle materials are summarized in Fig. 11. The diametral-compression test data show remarkably consistent trend despite the variety of different materials tested and their different microstructures. The inclined crack tests in uniaxial tension show more scatter, but their data trend is unmistakably different from the trend defined by the diametral-compression data. Results of Fig. 11 support the conclusion that there is no unique material-characteristic value of K_{IIC} for brittle materials. The test geometry and the stress state in the mixed-mode fracture test significantly affect the relative influence of mode II loading.

The second major conclusion involves the second order nonsingular stress term (and possibly higher order terms) in the series solutions of crack-tip stresses. Although quantitative agreement has not been achieved between the theoretical predictions and the results of the two types of tests, the rationale for the effect of the second order term appears to be reasonable. This rationale becomes particularly evident when we consider the limiting case of a crack oriented at $\alpha = 0$ degrees in the uniaxial test. Fracture criteria formulated in terms of only the mode I and mode II stress-intensity factors treat this limiting case as being equivalent to pure mode II loading (see Figs 6 and 7). It is clear from the angular dependence of stress-intensity factors for this test geometry that both K_I and K_{II} are zero at $\alpha = 0$ degrees. Fracture under this condition must be governed by stress contributions from terms other than the singular terms. Unfortunately, it is difficult to ascertain unambiguous values for the two critical parameters involved in the analysis employing the second order term; the near crack-tip location (r/a) where the fracture criterion is applied and the biaxial stress state parameter k for the diametral-compression test. A value of 0.01 was chosen arbitrarily for (r/a) for soda-lime glass. Williams and Ewing[3] and Finnie and Saith[4] also employed similar location parameters in their analysis of inclined crack test results in PMMA. For the inorganic soda-lime glass this location parameter might be expected to be smaller than that for PMMA simply on the basis of the relative ductility of the two materials. The value of -3 chosen for the stress state parameter k for the disk test is clearly an underestimate. This is based on the limiting ratio of the in-plane compressive stress and the transverse tensile stress in the center region of the disk specimen. Along the diametral line of loading this ratio increases quite significantly for radial locations away from the center. For disk specimens with finite crack sizes relative to the disk size, therefore, the second order stress term should be more significant. A closed form solution for this nonsingular term for the crack in the disk specimen is currently not available. It is interesting to note that with a higher numerical value for k for the disk specimen the theoretical prediction for $K_I - K_{II}$ envelope in Fig. 9 would be shifted toward higher K_{II} values and, therefore, the discrepancy between the theory and the disk test results would be reduced. Thus, underestimation of k for the disk specimen appears to be one major source of discrepancy between the experiments and the theory.

An appendix to this paper describes the results of calculations of the strain energy density criterion formulated with the second order stress terms retained in the expressions for the crack-tip stresses. The short-crack approximation and the values of the parameters (r/a) = 0.01 and $k = -3$ for the disk tests were also used in these calculations. Figures A1 and A2 compare the predictions of this modified strain energy density criterion with the disk and the tension test results. The qualitative trends of the predictions are similar to the predictions of the modified hoop stress criterion. There were, however, some interesting differences between the two. For example, the predictions for the disk test (curve labelled $k = -3$ in Fig. A2) does not give a better fit to the experimental data than does the original analysis based on K_I and K_{II} only (dashed line). It is, therefore, not clear that the strain energy density criterion can give any better predictions for the disk tests with more realistic values for the parameters (r/a) and k . On the whole, however, it is still difficult to choose among the theories because of the similarities of their predictions.

It is useful to compare the results of the present study with the through-cracks with a similar study conducted with surface cracks in the diametral-compression specimens[24]. Figure 12 compares the normalized mode I and mode II stress intensities for mixed-mode fracture from surface and through-cracks in the diametral-compression test. The surface cracks were introduced by indenting the specimens with a Knoop microhardness indenter. The mixed-mode fracture results with the surface cracks differed from the through-crack data in two ways. First, the surface cracks subjected to mixed-mode loading extended in directions normal to remote tension, i.e. the directions defined by eq. (3) (see also Fig. 6). Thus, the directions of extension of surface cracks deviate from

the theoretical predictions more than the corresponding extension directions for the through-cracks. The reason for the greater discrepancy with the surface cracks is not clear. Secondly, surface cracks show less sensitivity to mode II loading than the through-cracks. This is evidenced by the shift in the $K_I - K_{II}$ fracture envelope for the surface cracks in Fig. 12 to higher K_{II} values in relation to the through-crack data. It should be noted, however, that the through-crack data in Fig. 12 are for soda-lime glass while the surface crack data are for polycrystalline ceramics. Thus, the differences in the $K_I - K_{II}$ plots could be due to either the crack geometry or to the material microstructure. Generally, polycrystalline ceramics are likely to exhibit higher mode II fracture toughness than the theoretical expectations because surface asperities on the crack plane, for example, edges of grains in the case of intergranular cracks, resist crack-surface displacements generated by mode II loading. To separate the effects of crack geometry and the material microstructure experiments are currently planned where surface cracks and through-cracks will be compared with the same test material.

CONCLUSIONS

- (1) The center-cracked disk test in diametral compression provides a simple and effective method to study mixed-mode fracture of brittle materials.
- (2) The chevron notch enables stable precracking of the disk specimens to desired lengths.
- (3) The resulting through-cracks can be analyzed precisely for K_I and K_{II} by available solutions.
- (4) The results of the inclined crack fracture experiments with the disk specimens of soda-lime glass differ significantly from the inclined crack tests in uniaxial tension reported in the literature.
- (5) Mixed-mode fracture theories formulated only in terms of the mode I and the mode II stress-intensity factors do not satisfactorily account for the results of the disk tests.
- (6) The modifications of the theories by including the second order nonsingular terms in the expressions for the crack-tip stresses give better predictions for both types of tests. These analyses, however, suffer from the ambiguity in choosing values for the two critical parameters, the location parameter, (r/a) , and the stress state parameter, k .
- (7) It is difficult to discriminate among the different mixed-mode fracture criteria because of the similarity of their predictions.

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APPENDIX

Analysis of mixed-mode fracture by the modified strain energy density criterion

For plane strain condition, the elastic strain energy density for an isotropic elastic solid is given by the following relation[5]

$$\varphi = (\sigma_{xx}^2 + \sigma_{yy}^2 + \sigma_{zz}^2)/2E - \nu(\sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx})/E + (1 + \nu)\tau_{xy}^2/E. \quad (\text{A1})$$

For an inclined crack in a general biaxial stress state, as shown in Fig. 10, the expressions for the stress components σ_{xx} , σ_{yy} , σ_{zz} and τ_{xy} with the singular and the second order terms have been given by Eftis and Subramonian[27]. Only the stress component σ_{xx} has a nonsingular second order term. Substitution of these expressions into eq. (A1) results in the following equation for the strain energy density:

$$\varphi = a_{11}K_I^2/r + 2a_{12}K_I K_{II}/r + a_{22}K_{II}^2/r + bK_I\sigma/r^{1/2} - cK_{II}\sigma/r^{1/2} + d\sigma^2 \quad (\text{A2})$$

where the coefficients a_{11} , a_{12} and a_{22} are the same as those given in eqs (7), (8) and (9), respectively. The coefficients of the additional terms that arise from retaining the second order term of σ_{xx} are given by the following equations:

$$b = [(1-k) \cos 2\alpha/2G (2\pi)^{1/2}] \cos (\theta/2) \{1 - 2\nu - \sin (\theta/2) \sin (3\theta/2)\} \quad (\text{A3})$$

$$c = [(1-k) \cos 2\alpha/2G (2\pi)^{1/2}] \{2(1-\nu) \sin (\theta/2) + \sin (\theta/2) \cos (\theta/2) \cos (3\theta/2)\} \quad (\text{A4})$$

$$d = [(1-k)^2 \cos^2 2\alpha (1-\nu)]/4G. \quad (\text{A5})$$

The application of the modified strain energy density relation, eq. (A2), to analyze the mixed-mode fracture results involved using eq. 10 to define θ^* , the angle corresponding to the minimum value of φ (i.e. φ^*) assessed at a near crack-tip location (r/a). The resulting θ^* values for different crack-inclination angles were then used to calculate K_I and K_{II} combinations that gave the same minimum values of φ^* . The results obtained by this procedure are shown in Figs A1 and A2.

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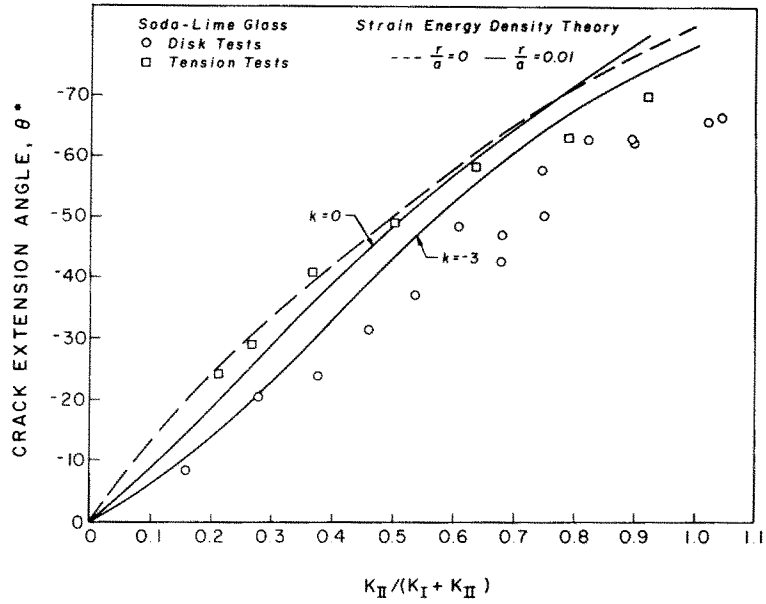


Fig. A1. Crack-extension angles for inclined cracks in diametral-compression and uniaxial tension tests as a function of the relative mode I and mode II stress intensities compared with the predictions of the modified strain energy density criterion (for $r/a = 0$ the result is independent of a).

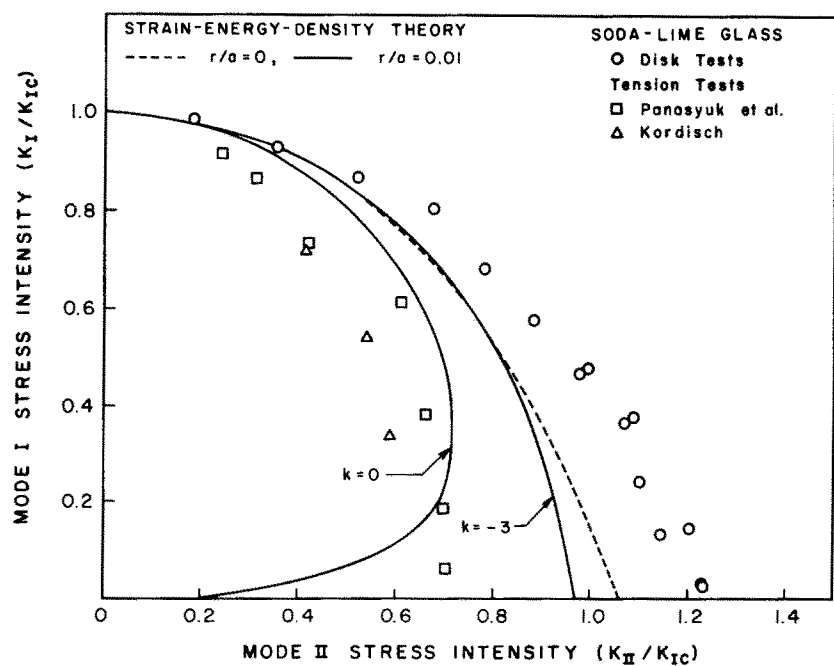


Fig. A2. Normalized mode I and mode II stress-intensity factors for mixed-mode fracture from inclined cracks in the disk and the uniaxial tests of soda-lime glass compared with the predictions of the modified strain energy density criterion (for $r/a = 0$ the result is independent of a).